

***** ==> REDUCE is over. Your final matrix is:

A =

1	0	0	$1/20$
0	1	0	$-3/5$
0	0	1	$3/2$

ans =

1.0000	0	0	0.0500
0	1.0000	0	-0.6000
0	0	1.0000	1.5000

>> rref()

Not enough input arguments.

Error in rref (line 23)

[m,n] = size(A);

>> a1 = [1 2 3 1; 1 3 0 1; 1 0 2 1]

a1 =

1	2	3	1
1	3	0	1
1	0	2	1

>> a2 = [8 7 3]'

a2 =

8
7
3

>> augA = [a1 a2]

augA =

1	2	3	1	8
1	3	0	1	7
1	0	2	1	3

>> rref(augA)

ans =

1	0	0	1	1
0	1	0	0	2
0	0	1	0	1

>> b1 = [1 1 3 -3; 0 2 1 -3; 1 0 2 -1]

```
b1 =
```

```
    1    1    3   -3
    0    2    1   -3
    1    0    2   -1
```

```
>> b2 = [0 3 -1]'
```

```
b2 =
```

```
    0
    3
   -1
```

```
>> augB = [b1 b2]
```

```
augB =
```

```
    1    1    3   -3    0
    0    2    1   -3    3
    1    0    2   -1   -1
```

```
>> rref([b1 b2])
```

```
ans =
```

```
    1    0    0    1    1
    0    1    0   -1    2
    0    0    1   -1   -1
```

```
>> rref(augB)
```

```
ans =
```

```
    1    0    0    1    1
    0    1    0   -1    2
    0    0    1   -1   -1
```

```
>> A = [1 2; -2 1]
```

```
A =
```

```
    1    2
   -2    1
```

```
>> rref(A)
```

```
ans =
```

```
    1    0
```

```
0      1
```

```
>> A = [1 2 3; 4 5 6; 7 8 9]
```

```
A =
```

```
1      2      3
4      5      6
7      8      9
```

```
>> rref(A)
```

```
ans =
```

```
1      0     -1
0      1      2
0      0      0
```

```
>> A = [1 2 3; 4 5 6; 7 8 0]
```

```
A =
```

```
1      2      3
4      5      6
7      8      0
```

```
>> rref(A)
```

```
ans =
```

```
1      0      0
0      1      0
0      0      1
```

```
>> A = [1 3; 1 2]
```

```
A =
```

```
1      3
1      2
```

```
>> rref([A eye(size(A))])
```

```
ans =
```

```
1      0     -2      3
0      1      1     -1
```

```
>> A = [1 1 2; 2 1 1; 1 2 1]
```

```
A =
```

```
1     1     2
2     1     1
1     2     1
```

```
>> rref([A eye(size(A))])
```

```
ans =
```

```
Columns 1 through 5
```

```
1.0000     0     0   -0.2500    0.7500
     0    1.0000     0   -0.2500   -0.2500
     0     0    1.0000    0.7500   -0.2500
```

```
Column 6
```

```
-0.2500
 0.7500
-0.2500
```

```
>> A = [1 2; 2 -1]
```

```
A =
```

```
1     2
2    -1
```

```
>> charpoly(A)
```

```
ans =
```

```
1     0    -5
```

```
>> A = [2 4 0; 1 2 1; 0 4 1]
```

```
A =
```

```
2     4     0
1     2     1
0     4     1
```

```
>> A = [2 4 0; 1 2 1; 0 4 2]
```

```
A =
```

```
2     4     0
1     2     1
0     4     2
```

```
>> charpoly(A)
```

ans =

1 -6 4 8

>> A = [1 0 0 0; 2 -2 0 0; 0 0 2 -1; 0 0 -1 2]

A =

1 0 0 0
2 -2 0 0
0 0 2 -1
0 0 -1 2

>> charpoly(A)

ans =

1 -3 -3 11 -6

>> A = [1 -3; 3 -5]

A =

1 -3
3 -5

>> e = eig(A)

e =

-2.0000
-2.0000

>> p = poly(e)

p =

1.0000 4.0000 4.0000

>> r = roots(p)

r =

-2.0000 + 0.0000i
-2.0000 - 0.0000i

>> A = [3 -1 4; -1 0 1; 4 1 2]

A =

```
    3    -1    4
   -1     0    1
    4     1    2
```

```
>> e = eig(A)
```

```
e =
```

```
 -2.3715
  0.8392
  6.5324
```

```
>> p = poly(e)
```

```
p =
```

```
 1.0000  -5.0000 -12.0000  13.0000
```

```
>> r = roots(p)
```

```
r =
```

```
 6.5324
 -2.3715
  0.8392
```

```
>> A = [2 -2 0; 1 -1 0; 1 -1 0]
```

```
A =
```

```
 2    -2     0
 1     -1     0
 1     -1     0
```

```
>> e = eig(A)
```

```
e =
```

```
 0
 0
 1
```

```
>> p = poly(e)
```

```
p =
```

```
 1    -1     0     0
```

```
>> r = roots(p)
```

```
r =
```

```
0
0
1

>> A = [2 4; 3 6]

A =

     2     4
     3     6

>> e = eig(A)

e =

     0
     8

>> p = poly(e)

p =

     1     -8     0

>> r = roots(p)

r =

     0
     8

>> load cities.mat
>> Cities = cities.mat
Unable to resolve the name cities.mat.

>> clear
>> load cities.mat
>> categories

categories =

9×14 char array

'climate      '
'housing      '
'health       '
'crime         '
'transportation'
'education     '
'arts          '
```

```
'recreation    '  
'economics    '
```

```
>> names
```

```
names =
```

```
329×43 char array
```

```
'Abilene, TX      '  
'Akron, OH       '  
'Albany, GA      '  
'Albany-Troy, NY '  
'Albuquerque, NM '  
'Alexandria, LA  '  
'Allentown,Bethlehem, PA-NJ  
'Alton, Granite City, IL '  
'Altoona, PA     '  
'Amarillo, TX    '  
'Anaheim-Santa Ana, CA '  
'Anchorage, AK   '  
'Anderson, IN    '  
'Anderson, SC    '  
'Ann Arbor, MI   '  
'Anniston, AL    '  
'Appleton-Oshkosh-Neenah, WI  
'Asheville, NC   '  
'Athens, GA      '  
'Atlanta, GA     '  
'Atlantic City, NJ '  
'Augusta, GA-SC  '  
'Aurora-Elgin, IL '  
'Austin, TX      '  
'Bakersfield, CA '  
'Baltimore, MD   '  
'Bangor, ME      '  
'Baton Rouge, LA '  
'Battle Creek, MI '  
'Beaumont-Port Arthur, TX  
'Beaver County, PA '  
'Bellingham, WA  '  
'Benton Harbor, MI '  
'Bergen-Passaic, NJ '  
'Billings, MT    '  
'Biloxi-Gulfport, MS  
'Binghampton, NY '  
'Birmingham, AL  '  
'Bismarck, ND     '  
'Bloomington, IN '  
'Bloomington-Normal, IL  
'Boise City, ID   '
```



```
'Boston, MA'
'Boulder-Longmont, CO'
'Bradenton, FL'
'Brazoria, TX'
'Bremerton, WA'
'Bridgeport-Milford, CT'
'Bristol, CT'
'Brockton, MA'
'Brownsville-Harlington, TX'
'Bryan-College Station, TX'
'Buffalo, NY'
'Burlington, NC'
'Burlington, VT'
'Canton, OH'
'Casper, WY'
'Cedar Rapids, IA'
'Champaign-Urbana-Rantoul, IL'
'Charleston, SC'
'Charleston, WV'
'Charlotte-Gastonia-Rock Hill, NC-SC'
'Charlottesville, VA'
'Chattanooga, TN-GA'
'Chicago, IL'
'Chico, CA'
'Cincinnati, OH-KY-IN'
'Clarksville-Hopkinsville, TN-KY'
'Cleveland, OH'
'Colorado Springs, CO'
'Columbia, MO'
'Columbia, SC'
'Columbus, GA-AL'
'Columbus, OH'
'Corpus Christi, TX'
'Cumberland, MD-WV'
'Dallas, TX'
'Danbury, CT'
'Danville, VA'
'Davenport-Rock Island-Moline, IA-IL'
'Dayton-Springfield, OH'
'Daytona Beach, FL'
'Decatur, IL'
'Denver, CO'
'Des Moines, IA'
'Detroit, MI'
'Dothgan, AL'
'Dubuque, IA'
'Duluth, MN-WI'
'East St. Louis-Bellefonte, IL'
'Eau Claire, WI'
'El Paso, TX'
'Elkhart-Goshen, IN'
```

```
'Elmira, NY'
'Enid, OK'
'Erie, PA'
'Eugene-Springfield, OR'
'Evansville, IN-KY'
'Fall River, MA-RI'
'Fargo-Moorhead, ND-MN'
'Fayetteville, NC'
'Fayetteville, AR'
'Fitchburg-Leominster, MA'
'Flint, MI'
'Florence, AL'
'Florence, SC'
'Fort Collins-Loveland, CO'
'Fort Lauderdale-Hollywood-Pompano Beach, FL'
'Fort Myers, FL'
'Fort Pierce, FL'
'Fort Smith, AR-OK'
'Fort Walton Beach, FL'
'Fort Wayne, IN'
'Fort Worth-Arlington, TX'
'Fresno, CA'
'Gadsden, AL'
'Gainesville, FL'
'Galveston-Texas City, TX'
'Gary-Hammond, IN'
'Glens Falls, NY'
'Grand Forks, ND'
'Grand Rapids, MI'
'Great Falls, MT'
'Greeley, CO'
'Green Bay, WI'
'Greensboro-Winston-Salem-High Point, NC'
'Greenville-Spartanburg, SC'
'Hagerstown, MD'
'Hamilton-Middletown, OH'
'Harrisburg-Lebanon-Carlisle, PA'
'Hartford, CT'
'Hickory, NC'
'Honolulu, HI'
'Houma-Thibodaux, LA'
'Houston, TX'
'Huntington-Ashland, WV-KY-OH'
'Huntsville, AL'
'Indianapolis, IN'
'Iowa City, IA'
'Jackson, MI'
'Jackson, MS'
'Jacksonville, FL'
'Jacksonville, NC'
'Janesville-Beloit, WI'
```

```
'Jersey City, NJ'
'Johnson City-Kingsport-Bristol, TN-VA'
'Johnstown, PA'
'Joliet, IL'
'Joplin, MO'
'Kalamazoo, MI'
'Kankakee, IL'
'Kansas City, KS'
'Kansas City, MO'
'Kenosha, WI'
'Killeen-Temple, TX'
'Knoxville, TN'
'Kokomo, IN'
'La Crosse, WI'
'Lafayette, IN'
'Lafayette, LA'
'Lake Charles, LA'
'Lake County, IL'
'Lakeland-Winter Haven, FL'
'Lancaster, PA'
'Lansing-East Lansing, MI'
'Laredo, TX'
'Las Cruces, NM'
'Las Vegas, NV'
'Lawrence, KS'
'Lawrence-Haverhill, MA-NH'
'Lawton, OK'
'Lewiston-Auburn, ME'
'Lexington-Fayette, KY'
'Lima, OH'
'Lincoln, NE'
'Little Rock, North Little Rock, AR'
'Longview-Marshall, TX'
'Lorain-Elyria, OH'
'Los Angeles, Long Beach, CA'
'Louisville, KY-IN'
'Lowell, MA-NH'
'Lubbock, TX'
'Lynchburg, VA'
'Macon, Warner Robbins, GA'
'Madison, WI'
'Manchester, NH'
'Mansfield, OH'
'McAllen-Edinburg-Mission, TX'
'Medford, OR'
'Melbourne-Titusville-Palm Bay, FL'
'Memphis, TN-AR-MS'
'Miami-Hialeah, FL'
'Middlesex-Somerset, Hunterdon, NJ'
'Middletown, CT'
'Midland, TX'
```

```
'Milwaukee, WI'
'Minneapolis-St. Paul, MN-WI'
'Mobile, AL'
'Modesto, CA'
'Monmouth-Ocean, NJ'
'Monroe, LA'
'Montgomery, AL'
'Muncie, IN'
'Muskegon, MI'
'Nashua, NH'
'Nashville, TN'
'Nassua-Suffolk, NY'
'New Bedford, MA'
'New Britain, CT'
'New Haven-Meriden, CT'
'New London-Norwich, CT-RI'
'New Orleans, LA'
'New York, NY'
'Newark, NJ'
'Niagara Falls, NY'
'Norfolk-Virginia Beach-Newport News, VA'
'Norwalk, CT'
'Oakland, CA'
'Ocala, FL'
'Odessa, TX'
'Oklahoma City, OK'
'Olympia, WA'
'Omaha, NE-IA'
'Orange County, NY'
'Orlando, FL'
'Owensboro, KY'
'Oxnard-Ventura, CA'
'Panama City, FL'
'Parkerburg-Marietta, WV-OH'
'Pascagoula, MS'
'Pawtucket-Woonsocket-Attleboro, RI-MA'
'Pensacola, FL'
'Peoria, IL'
'Philadelphia, PA-NJ'
'Phoenix, AZ'
'Pine Bluff, AR'
'Pittsburgh, PA'
'Pittsfield, MA'
'Portland, ME'
'Portland, OR'
'Portsmouth-Dover-Rochester, NH-ME'
'Poughkeepsie, NY'
'Providence, RI'
'Provo-Orem, UT'
'Pueblo, CO'
'Racine, WI'
```

'Raleigh-Durham, NC	'
'Reading, PA	'
'Redding, CA	'
'Reno, NV	'
'Richland-Kinnewick-Pasco, WA	'
'Richmond-Petersburg, VA	'
'Riverside-San Bernardino, CA	'
'Roanoke, VA	'
'Rochester, MN	'
'Rochester, NY	'
'Rockford, IL	'
'Sacramento, CA	'
'Saginaw-Bay City-Midland, MI	'
'St. Cloud, MN	'
'St. Joseph, MO	'
'St. Louis, MO-IL	'
'Salem, OR	'
'Salem, MA	'
'Salinas-Seaside-Monterey, CA	'
'Salt Lake City-Ogden, UT	'
'San Angelo, TX	'
'San Antonio, TX	'
'San Diego, CA	'
'San Francisco, CA	'
'San Jose, CA	'
'Santa Barbara-Santa Maria-Lompoc, CA	'
'Santa Cruz, CA	'
'Santa Rosa-Petaluma, CA	'
'Sarasota, FL	'
'Savannah, GA	'
'Scranton-Wilkes Barre, PA	'
'Seattle, WA	'
'Sharon, PA	'
'Sheboygan, WI	'
'Sherman-Denison, TX	'
'Shreveport, LA	'
'Sioux City, IA-NE	'
'Sioux Falls, SD	'
'South Bend-Mishawaka, IN	'
'Spokane, WA	'
'Springfield, IL	'
'Springfield, MA	'
'Springfield, MO	'
'Stamford, CT	'
'State College, PA	'
'Steubenville-Weirton, OH-WV	'
'Stockton, CA	'
'Syracuse, NY	'
'Tacoma, WA	'
'Tallahassee, FL	'
'Tampa-St. Petersburg-Clearwater, FL	'

```

'Terre Haute, IN
'Texarkana, TX-Texarkana, AR
'Toledo, OH
'Topeka, KS
'Trenton, NJ
'Tuscon, AZ
'Tulsa, OK
'Tuscaloosa, AL
'Tyler, TX
'Utica-Rome, NY
'Vallejo-Fairfield-Napa, CA
'Vancouver, WA
'Victoria, TX
'Vineland-Millville-Bridgeton, NJ
'Visalia-Tulare-Porterville, CA
'Waco, TX
'Washington, DC-MD-VA
'Materbury, CT
'Waterloo-Cedar Falls, IA
'Wausau, WI
'West Palm Beach-Boca Raton-Delray Beach, FL'
'Wheeling, WV-OH
'Wichita, KS
'Wichita Falls, TX
'Williamsport, PA
'Wilmington, DE-NJ-MD
'Wilmington, NC
'Worcester, MA
'Yakima, WA
'York, PA
'Youngstown-Warren, OH
'Yuba City, CA

```

```
>> ratings
```

```
ratings =
```

```
Columns 1 through 4
```

521	6200	237	923
575	8138	1656	886
468	7339	618	970
476	7908	1431	610
659	8393	1853	1483
520	5819	640	727
559	8288	621	514
537	6487	965	706
561	6191	432	399
609	6546	669	1073
885	16047	2025	983
195	12175	601	1223

530	5704	580	878
591	5725	820	975
546	11014	2508	1067
560	5530	598	1125
396	7877	833	525
694	6722	1204	566
601	6691	605	933
696	8316	3195	1308
615	11074	637	1878
534	6292	1798	872
474	10384	1203	821
435	8831	782	1049
560	8068	420	1561
567	9148	3562	1730
451	7277	780	651
427	8083	342	1565
527	6342	900	1031
423	6288	616	1313
586	7866	861	310
772	8329	240	825
566	6761	570	1190
559	14607	2661	857
452	8315	479	810
584	6458	441	810
550	8257	1007	415
612	6811	1692	1123
149	8365	804	413
558	7056	731	657
487	8654	815	673
592	8221	453	880
623	11609	5301	1215
459	11914	962	1088
440	8242	333	1093
423	8394	438	768
808	9060	310	651
648	13429	2550	943
516	10041	975	545
575	8263	916	1336
440	5376	91	974
383	8228	640	1016
571	8064	2465	971
637	6179	994	707
383	9673	1809	494
575	7332	443	650
401	9839	345	989
434	7774	837	714
525	8627	672	1022
569	7402	1463	1495
627	7789	708	721
644	7169	999	1273
618	9531	1348	756

576	6189	564	946
514	10913	5766	1034
603	8587	243	947
584	8143	2138	978
544	6007	446	736
579	9168	3167	1138
526	8509	721	1086
541	7702	1951	1065
526	7519	1421	1524
517	5817	833	820
558	8093	1837	1092
362	6929	458	1335
591	6054	760	337
544	9318	2825	1529
569	14420	2350	548
545	5709	593	379
440	8083	1113	834
544	7635	2253	1151
561	7203	723	1347
480	7395	732	897
521	10789	2533	1365
444	8028	1256	1044
536	8525	4142	1587
336	5708	593	930
419	7993	640	571
193	6040	1159	488
537	6501	444	1096
257	7078	798	433
592	7343	528	1323
521	6573	596	524
467	7078	562	582
461	6829	626	845
605	7715	529	635
741	9370	539	874
550	6743	783	864
643	9017	900	861
148	8168	920	503
561	6274	872	1150
549	6686	594	545
507	8252	655	655
540	7204	724	1671
546	5962	607	516
552	6508	818	1334
490	9951	731	744
572	10810	1252	1536
342	9298	546	787
602	8842	527	1422
482	5784	466	663
536	7554	484	544
509	6733	1060	710
528	7956	1038	1348

559	9291	369	1483
526	5382	622	749
402	7388	1731	1658
727	7767	1437	1213
483	7641	1364	996
476	7120	43	568
105	7898	1109	401
513	7780	1274	952
410	7143	667	792
490	8218	706	994
367	8401	916	583
626	7064	1694	967
655	6336	1260	1185
568	7763	818	627
542	8227	1135	892
556	7891	2087	629
516	11652	2521	1279
623	6760	1006	765
717	17021	1298	891
427	7094	583	400
424	9760	2467	1499
636	6632	875	665
600	6283	685	924
557	7012	2243	1000
434	9429	2437	830
518	6794	679	1021
412	7245	1792	1091
457	6626	1181	1211
564	6111	740	967
466	7447	700	858
601	8810	1759	1434
663	6119	1152	424
547	6524	731	353
479	9327	1058	837
580	5159	500	628
527	7919	1043	1120
483	7230	609	976
549	8126	1711	1142
549	7076	1939	1468
496	8516	1067	911
365	6463	398	733
670	6692	960	622
512	6616	596	413
352	8310	686	676
494	7778	655	465
429	8572	548	1030
469	6921	314	1093
514	13282	1237	822
307	6680	323	1373
559	8631	1111	413
480	7907	1371	894

424	6152	465	1050
552	6962	588	1457
556	9906	412	1913
513	7497	621	1018
548	10414	1202	909
479	5850	477	1156
490	6876	759	764
635	8340	1860	1055
522	6986	741	855
398	8256	775	789
497	7270	1861	1328
500	6608	509	976
579	8309	1105	609
885	13868	5153	1960
616	6812	2111	937
526	9640	1083	819
604	6990	900	1608
642	6934	732	643
447	6235	593	783
378	9897	2168	779
404	9860	737	633
558	6881	303	1072
238	5345	372	836
611	9008	256	728
582	8721	517	1039
514	7015	2043	1488
634	10267	2314	2459
559	12135	2589	691
593	11652	884	646
603	8672	97	1166
460	10176	3053	826
293	9559	3934	906
442	6704	1469	1511
639	8630	347	1154
615	11660	2482	819
455	6190	331	957
483	6754	832	815
530	5800	949	783
580	6391	699	1537
538	10757	853	452
600	7800	1850	984
656	11138	3919	566
643	8087	519	1012
516	10509	1245	903
583	11460	2068	893
583	10218	556	633
498	8515	2586	1604
638	13358	7850	2498
601	14220	4106	1461
554	7686	507	775
632	8568	1932	997

648	20151	2530	625
910	13135	2362	1533
333	6750	489	1327
603	6689	384	1698
554	7186	1623	1297
726	8263	338	752
440	7128	2559	1008
509	10173	1574	804
457	8196	765	1671
524	6760	812	568
890	14000	1106	791
536	6373	201	1344
617	6657	665	488
584	6248	593	591
586	9462	1117	744
536	6479	563	1472
491	8388	1184	921
630	8310	5158	1059
536	8921	1584	1268
463	5674	617	1169
586	8099	3413	687
482	7807	694	638
483	8100	834	823
768	9912	1590	1504
469	9966	596	475
488	9981	355	633
586	9274	2467	998
500	9321	198	485
497	6637	468	1181
496	8943	931	1055
647	8230	3476	981
614	7614	1154	522
664	8584	274	892
535	12449	615	1116
664	8461	300	779
585	8343	2448	1076
615	9754	2201	1475
652	7476	1036	784
308	9193	2966	437
536	8609	1969	894
466	7584	969	1156
576	9855	1027	1363
515	7368	1022	1068
195	7235	603	343
475	5589	223	969
537	7605	2850	1306
716	8378	749	1014
644	11622	1232	490
843	13838	352	1107
541	9466	1631	969
488	6321	236	1032

398	6898	1337	1197
903	14465	2416	1099
910	17158	3726	1619
850	16048	2117	1065
855	15547	532	1026
843	14303	1035	964
732	12931	1052	912
391	9560	801	939
542	6896	1084	1614
575	6697	1219	372
808	10183	2715	1170
570	6697	700	384
442	8121	593	450
524	5722	394	1035
508	6534	1445	1197
385	6528	846	759
276	7983	1041	556
545	5938	830	1038
574	6927	497	869
524	7882	1877	1225
453	8039	710	1212
544	6343	577	892
648	23640	2610	835
575	8405	612	540
542	6578	505	418
625	8474	342	1395
548	7670	1040	689
808	7770	539	1162
404	8029	370	1161
440	7442	1189	1493
557	5527	453	630
467	5717	343	822
518	7767	1738	998
501	7110	1148	999
636	10616	1372	1181
589	8548	1259	1400
530	7498	1581	1080
470	6464	674	1014
500	7298	672	955
548	6744	391	400
821	10503	1079	964
768	9015	517	752
336	7143	260	1092
615	7295	807	1135
543	7778	210	1132
412	6106	538	1166
631	13724	4361	1317
569	10024	1218	789
347	7881	925	700
308	7642	818	442
509	10512	375	1783

542	6576	791	308
494	7061	806	1164
456	6404	549	1179
558	7284	860	464
597	7927	1445	1115
564	6858	1099	1423
562	8715	1805	680
535	6440	317	1106
540	8371	713	440
570	7021	1097	938
608	7875	212	1179

Columns 5 through 8

4031	2757	996	1405
4883	2438	5564	2632
2531	2560	237	859
6883	3399	4655	1617
6558	3026	4496	2612
2444	2972	334	1018
2881	3144	2333	1117
4975	2945	1487	1280
4246	2778	256	1210
4902	2852	1235	1109
3954	2843	5632	3156
5091	2414	2346	3000
2865	2469	430	838
2707	2772	169	613
3433	3346	7559	2288
3051	2189	268	1165
3298	2844	1166	2315
5086	2990	1391	1542
1866	2646	3546	1001
8409	3057	7559	1362
3556	2929	621	2711
2523	2915	1047	913
3943	2208	3857	1800
3670	3063	5355	1063
3725	2564	1222	1568
7405	3471	9788	2925
5613	2934	1995	2148
3329	2635	4237	1413
4652	2483	354	1648
2782	2745	1795	1813
2960	2535	1284	1480
3776	2778	1302	3200
2989	2545	79	1477
3511	3653	9304	1918
6285	3008	778	2046
2516	2592	679	2106
4529	3052	1599	1722

5177	2851	3958	1234
4303	2686	1211	1630
1746	2873	2152	1990
5889	2854	1470	1605
6575	2391	2385	1672
6801	3479	21042	3066
7108	2587	3663	4012
3805	2712	154	1349
2391	2718	1506	1512
1670	2544	382	1973
3197	3029	8368	1913
4495	2628	514	777
3810	2729	2001	1217
3119	2413	162	3000
2530	2973	2002	1413
5384	3121	8567	2441
1910	2519	131	701
7146	3323	1741	3357
4279	2754	989	1157
4410	2453	303	1435
5270	2619	904	1501
7447	3147	2203	1700
4207	3164	2993	2561
5470	2894	2605	844
6099	3031	4313	1236
6041	3489	1422	1704
3401	2415	2483	1238
7742	3486	24846	2856
4067	3126	1647	1543
5748	2918	9688	2451
2226	2654	111	2219
7333	2972	12679	3300
3389	2754	1749	2375
3893	2377	2882	1331
5859	2908	2489	1484
2995	2665	1861	1214
4364	2928	6648	2020
3626	2840	1992	2037
3709	3363	373	1036
6213	3269	10438	2310
2715	3029	7415	1572
3161	2943	85	501
3907	2901	1017	1920
4775	2772	6935	1122
4117	2612	809	3967
3867	2683	298	1222
8145	3145	8477	2324
5521	2613	1857	1802
4808	3064	10389	2483
2232	2230	117	714
3668	2701	340	1587

5205	2619	2377	3107
6539	2630	904	1610
3197	2960	1807	1397
3705	2479	3800	1101
4168	2537	353	1023
3324	3000	1048	1600
2312	2764	215	1200
5754	2641	2032	1340
5293	3118	2631	3400
3496	2797	1876	1622
4602	2439	749	2005
6325	2506	2111	1414
4402	3051	844	709
3581	2334	1915	1695
2244	2799	270	790
4912	2511	2163	1355
1454	2427	1021	994
4963	3109	628	800
2637	2413	1609	4200
4186	2734	2027	2455
4583	2729	380	4005
2143	3154	368	2058
3092	2927	145	1736
2886	2809	87	2092
5416	2772	2846	1711
4472	2627	6466	2366
4388	2407	3596	2984
2174	2299	153	300
3527	3094	3335	1237
3423	2809	1756	3000
5855	2526	4115	1940
2241	2674	603	1883
5587	2721	1921	1304
3454	2705	3255	1909
3747	2737	401	2176
1641	2854	1254	739
3793	2622	2547	1925
4453	3090	4188	1651
3950	3236	2569	1410
3431	2990	825	1491
3338	2747	2316	1604
6164	3224	3083	1532
7120	3628	3616	1790
2703	2726	188	797
5911	2502	7168	3703
1145	1995	725	2700
4626	3271	11073	1825
4001	2525	2195	840
2661	2257	1921	1075
5804	2690	6348	1906
2770	2842	2255	1506

4800	2654	323	1933
4917	3130	3209	1427
5611	3048	2162	2884
1780	2646	567	1177
3092	2532	1092	1615
4982	2574	7420	1001
2532	2925	1925	2155
4343	2691	666	903
4645	2868	3177	1636
2335	2871	80	801
5419	2896	2071	2163
3444	2855	75	1119
5006	3028	1167	1045
5869	2949	5553	2043
4473	2918	3844	2224
2862	3167	920	931
4273	2761	3309	2514
1817	2904	285	1000
6096	3027	1466	1953
4956	2945	2235	1814
5268	2305	1772	1734
3549	2336	1456	1855
3422	2607	3746	2435
3412	2998	309	2513
3908	3097	1015	1147
5557	2891	4206	1609
3322	2827	150	702
2989	2736	2804	1609
5900	2241	1586	3996
2931	2700	3150	1752
3575	2479	2111	1879
2366	2375	1280	1757
2941	2694	736	1853
4080	2861	3596	1403
4084	2629	1352	1428
5618	2878	4523	1804
4186	2581	2180	1462
2680	2816	334	834
3629	2582	2565	1602
4345	3195	23567	3948
5420	3028	4916	1942
3820	2479	3057	1129
4158	2545	3402	1702
4909	2803	1079	1439
3144	2651	1435	1204
6084	3047	5123	1944
4595	2728	475	837
2876	2871	554	1133
2117	2644	1231	1059
3512	2797	1856	1416
2560	2814	437	3800

6247	2804	4486	1994
5202	2879	4837	4300
4198	3539	8058	1596
4636	3128	730	1682
5310	2416	438	1502
4945	3044	8766	2902
5606	3013	11714	2158
3345	2779	1764	2164
2000	2616	631	833
2690	2787	7563	3544
3606	2453	1528	1541
3509	2388	1374	1112
4325	2965	2498	1428
3353	2630	529	2666
3320	2728	2122	1523
5030	2763	4342	1849
2119	3234	8640	3705
3219	2439	766	1450
4900	3128	1360	1217
5938	3495	7852	1604
4505	3244	1164	2281
4579	2995	7978	3500
8625	2984	56745	3579
3514	3362	14224	1818
5561	2538	966	1873
3215	3014	7087	2964
3536	3029	7273	2268
6430	2646	6162	2394
3798	2864	266	3095
1944	2721	259	900
4459	2908	4843	1742
4083	2625	708	1451
5806	3069	3787	1977
3299	2754	6987	1866
5887	2976	2681	2881
2273	2491	804	852
2238	2155	2769	2135
2778	2500	755	2089
4399	2503	91	1148
1750	1701	155	1956
4738	3058	480	1513
2918	2914	1954	2160
2967	2452	2294	1688
5903	3781	17270	1979
4729	2942	4573	2472
1671	2554	373	793
5616	3544	11069	2145
3759	3264	228	1420
5185	2973	2351	3366
5947	3343	5160	2532
2321	3026	165	1390

4166	2898	785	1670
5474	3558	6152	2263
4546	2618	1985	3300
3501	2653	1307	1619
3558	2732	1171	2016
6544	3455	5730	1606
3120	3028	1108	1549
5727	2471	845	2424
6767	2529	2210	2386
4714	1728	529	1204
6680	2940	5697	1943
3141	2596	5327	1918
3872	2723	1263	2036
4399	2134	769	1503
5165	3582	6956	2659
2987	2680	1026	933
5097	2793	4483	2306
3186	2772	1708	2059
4565	2502	1871	1572
2689	2927	879	1265
7119	3530	8896	2243
4732	3278	691	1873
3459	2729	3276	2234
4160	2439	1004	3179
6228	2340	5528	3900
2938	2707	1301	1136
5387	2938	4295	1509
5489	2794	8818	3347
8299	3371	14226	4600
5224	2709	6446	1964
5662	2719	2684	3300
5010	2611	3748	1703
3313	2722	3457	2255
3742	2626	817	2535
5958	2456	2262	2237
3683	3230	1832	1386
6634	2710	9577	4800
2017	3022	52	1100
3458	2557	268	1316
1922	2652	68	937
4401	2858	2826	1389
4316	2673	1393	1359
6271	2651	465	1324
5634	2874	2672	1819
5534	2774	2988	1517
6172	3078	1983	1536
6159	3525	3466	1514
3828	2709	1634	1737
3110	3029	7865	1729
2740	3169	1271	1200
1532	2672	147	1460

4427	2155	1579	1630
6951	3144	5080	2851
4730	2546	4297	4000
5530	2790	2181	1936
5588	3044	5040	2943
3550	3012	1226	1401
2537	2899	63	669
5323	2852	4389	1952
5348	2795	1632	1141
4786	3311	5029	1646
4397	2685	4889	3131
3758	2628	4248	2024
4723	2390	1432	1090
3460	3283	404	631
4592	2970	858	1750
4153	2498	2962	1559
3817	2332	1557	1464
2407	2696	87	1410
4133	2747	2097	1474
3094	2128	511	2800
3018	2867	1141	1248
8236	3635	21701	1578
2434	2995	318	946
3351	2889	3000	1900
3496	2749	761	1654
5201	3224	2888	3772
2450	3002	422	1271
3933	2981	2987	1508
2793	2747	599	1126
3097	2906	196	726
4532	3112	4545	1923
2904	2876	1077	2668
3643	3299	1784	910
3731	2491	996	2140
2267	2903	1022	842
3374	2920	2797	1327
2768	2387	122	918

Column 9

7633
4350
5250
5864
5727
5254
5097
5795
4230
6241
6220

7668
3370
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4579
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8107
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7439
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4008
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4843
6019
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5671

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5563
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6072
5656
4407
4300

7992
4740
6036
6805
3288
6174
5390
5040
4986
4946
3894
4694

```
>> CovCategories = cov(categories)
```

```
CovCategories =
```

```
1.0e+03 *
```

```
Columns 1 through 5
```

0.0451	-0.0025	-0.0356	0.0060	0.0812
-0.0025	0.0436	0.0029	0.0140	-0.0648
-0.0356	0.0029	0.0711	-0.0029	-0.1074
0.0060	0.0140	-0.0029	0.0241	-0.0388
0.0812	-0.0648	-0.1074	-0.0388	0.6502
0.0959	-0.1263	-0.0425	-0.0553	0.5813
0.1319	-0.0804	0.0103	-0.0164	0.4060
0.1637	-0.1149	-0.0474	-0.0679	0.3299
0.1689	-0.1325	-0.0495	-0.0632	0.3647
0.1970	0.0004	-0.1612	0.0410	0.1813
0.1272	0.0747	-0.1062	0.0012	0.1867
0.1105	0.0649	-0.0923	0.0010	0.1622
0.1196	0.0702	-0.0998	0.0011	0.1756
0.1181	0.0693	-0.0986	0.0011	0.1733

```
Columns 6 through 10
```

0.0959	0.1319	0.1637	0.1689	0.1970
-0.1263	-0.0804	-0.1149	-0.1325	0.0004
-0.0425	0.0103	-0.0474	-0.0495	-0.1612
-0.0553	-0.0164	-0.0679	-0.0632	0.0410
0.5813	0.4060	0.3299	0.3647	0.1813
1.1918	0.9909	0.6300	0.6697	0.2112
0.9909	1.4196	1.0314	1.1076	0.5691
0.6300	1.0314	1.5895	1.6927	0.7802
0.6697	1.1076	1.6927	1.8257	0.8093
0.2112	0.5691	0.7802	0.8093	1.0046
0.2100	0.3057	0.5098	0.5040	0.5157
0.1825	0.2656	0.4431	0.4380	0.4481
0.1975	0.2875	0.4795	0.4740	0.4850

```
0.1950    0.2838    0.4734    0.4680    0.4788
```

```
Columns 11 through 14
```

```
0.1272    0.1105    0.1196    0.1181
0.0747    0.0649    0.0702    0.0693
-0.1062   -0.0923   -0.0998   -0.0986
0.0012    0.0010    0.0011    0.0011
0.1867    0.1622    0.1756    0.1733
0.2100    0.1825    0.1975    0.1950
0.3057    0.2656    0.2875    0.2838
0.5098    0.4431    0.4795    0.4734
0.5040    0.4380    0.4740    0.4680
0.5157    0.4481    0.4850    0.4788
0.7840    0.6813    0.7373    0.7280
0.6813    0.5921    0.6408    0.6327
0.7373    0.6408    0.6934    0.6847
0.7280    0.6327    0.6847    0.6760
```

```
>> categories
```

```
categories =
```

```
9x14 char array
```

```
'climate      '
'housing      '
'health       '
'crime        '
'transportation'
'education    '
'arts         '
'recreation   '
'economics    '
```

```
>> CovNames = cov(names)
```

```
CovNames =
```

```
Columns 1 through 5
```

```
48.5552   -7.5353   -0.9695  -21.0835  -11.9076
-7.5353   50.1236  -10.5791   13.3716    4.9793
-0.9695  -10.5791  121.3286   30.2656  -11.0714
-21.0835   13.3716   30.2656  378.7792  114.9619
-11.9076    4.9793  -11.0714  114.9619  441.2899
-20.0896   -5.3258   -9.6806  -66.9414  147.2014
 3.8728   -1.7643  -29.8326  -92.5249  -72.9181
-1.7971    1.8923   18.3098  -83.7020 -124.3648
 0.6702   -8.8991   22.6984  -67.7262  -88.0475
-12.6895   -9.5825   27.3048   13.5834   -6.8291
```

-23.4895	-21.5170	22.3528	32.1847	31.5135
-3.3008	-15.5745	8.1683	6.1181	27.7702
2.4602	-10.4832	11.4819	-23.1268	-9.4155
2.1322	-15.5996	47.8766	14.9667	-53.7955
-3.1605	-19.6208	30.6565	12.0646	-20.5559
-3.2361	-6.9488	22.8474	22.3021	-12.2069
4.3248	-13.5855	43.0276	33.4650	-16.3095
2.1384	-21.1065	34.8144	66.3516	-3.5064
10.5493	-22.4825	23.9720	57.9017	-15.3408
8.9645	-16.0879	24.0379	38.0843	-30.7967
-0.0005	-6.8996	16.6554	11.0499	-7.4633
5.3103	-6.5524	8.5245	15.5023	4.0216
12.8935	-19.3562	9.9961	32.7887	-23.7605
15.4054	-19.3834	13.6788	33.7886	-31.0623
7.5838	-9.4126	21.3475	24.0167	-18.4964
8.5833	-1.4920	13.3261	16.4782	5.2872
7.1837	-4.4457	-0.3408	17.2504	18.3933
5.3908	-7.4433	-1.5588	24.5528	-0.1194
8.4062	-4.0759	9.8031	20.0336	-17.5862
7.8771	-3.9423	11.4778	22.8013	-16.6961
3.9343	-4.9660	3.6899	16.9354	-23.3589
3.7509	-0.0355	3.6759	9.7803	-19.2150
3.3790	1.6065	6.1042	11.9047	-19.8599
-0.5265	0.6061	2.7007	6.5348	12.7902
0.9532	-0.0136	2.9181	6.4012	-6.1093
2.8085	-0.5472	4.4883	9.7290	-23.4096
1.8205	-0.2063	3.7727	7.2482	-23.7771
1.3997	2.1793	3.2492	4.9656	-24.6213
1.4450	1.5689	3.2428	5.4837	-27.7290
0.2599	0.0240	0.5359	0.9615	-4.9812
0	0	0	0	0
0.8229	0.0761	1.6970	3.0446	-15.7737
0.9529	0.0881	1.9649	3.5253	-18.2643

Columns 6 through 10

-20.0896	3.8728	-1.7971	0.6702	-12.6895
-5.3258	-1.7643	1.8923	-8.8991	-9.5825
-9.6806	-29.8326	18.3098	22.6984	27.3048
-66.9414	-92.5249	-83.7020	-67.7262	13.5834
147.2014	-72.9181	-124.3648	-88.0475	-6.8291
602.0634	217.3730	-99.3707	-111.2614	-5.3834
217.3730	778.7214	389.7871	-101.9157	-94.4574
-99.3707	389.7871	955.3841	468.3853	-20.6709
-111.2614	-101.9157	468.3853	991.1835	414.9600
-5.3834	-94.4574	-20.6709	414.9600	895.0974
17.8959	109.7792	66.8593	-81.8023	362.2500
32.0728	137.4183	309.6298	143.3968	-56.9470
19.4265	156.7925	341.4632	377.6952	217.4687
-22.0417	163.0539	315.5881	338.7947	412.2333
-29.3656	123.2793	266.8491	283.5328	291.9862

-4.8942	99.2346	189.9019	218.6513	236.9630
-1.0892	60.8098	166.7508	189.2589	198.2829
-24.9237	79.9307	164.1513	146.9302	127.0660
-38.1016	109.2277	114.7231	83.1204	80.0098
-5.0967	135.9970	118.6813	85.7572	59.7518
6.0708	87.1615	69.6400	77.4081	103.4084
21.2864	75.3542	54.1597	79.5750	71.0167
-9.7468	83.1912	92.9578	75.6352	-2.0931
-21.2544	79.3613	76.6057	69.4344	18.1795
-9.6298	49.1986	55.0861	62.6169	25.0805
17.4044	62.3083	53.7841	53.9184	2.1200
8.0075	54.3983	44.1296	44.6238	3.2766
-8.3791	39.6433	21.2919	41.2353	-8.9991
-19.7636	17.0731	17.3366	40.0373	-7.3550
-22.5145	1.7740	10.9588	36.9263	10.8777
-26.2057	8.5157	42.3737	25.6168	-9.6032
-11.7772	15.9529	33.3927	35.4108	-2.5032
1.2004	15.6782	16.1710	39.2343	-1.3378
6.8398	6.2430	1.2111	17.8190	2.4459
-7.4897	9.3096	12.9154	19.3600	-0.1386
-11.7161	5.6316	5.3888	15.8903	-1.2855
-6.4693	5.5379	6.9462	10.6240	-5.5430
-4.0817	5.6356	8.0203	14.0934	3.9196
-5.9073	4.5999	10.5737	13.4731	0.7765
-1.4574	0.3948	1.9839	1.6212	-0.8882
0	0	0	0	0
-4.6151	1.2501	6.2825	5.1338	-2.8125
-5.3438	1.4475	7.2745	5.9444	-3.2566

Columns 11 through 15

-23.4895	-3.3008	2.4602	2.1322	-3.1605
-21.5170	-15.5745	-10.4832	-15.5996	-19.6208
22.3528	8.1683	11.4819	47.8766	30.6565
32.1847	6.1181	-23.1268	14.9667	12.0646
31.5135	27.7702	-9.4155	-53.7955	-20.5559
17.8959	32.0728	19.4265	-22.0417	-29.3656
109.7792	137.4183	156.7925	163.0539	123.2793
66.8593	309.6298	341.4632	315.5881	266.8491
-81.8023	143.3968	377.6952	338.7947	283.5328
362.2500	-56.9470	217.4687	412.2333	291.9862
968.9277	476.3994	145.0222	352.0784	481.5623
476.3994	991.6175	601.0289	283.0683	441.5171
145.0222	601.0289	992.5975	623.4361	399.2991
352.0784	283.0683	623.4361	950.3663	660.1454
481.5623	441.5171	399.2991	660.1454	918.2775
323.8476	500.0919	542.0432	473.6297	643.0014
275.7831	455.5760	601.0411	549.5753	509.4047
288.9016	458.9728	534.4162	589.4463	589.2027
240.1899	352.8844	413.2673	444.4121	470.7377
217.5697	318.9440	381.0693	419.4603	464.4739

201.5884	273.0499	325.6485	356.0747	434.4923
149.6577	233.2335	290.1750	344.5104	408.0014
140.3750	233.4603	288.6354	335.0294	354.3731
147.6868	250.0523	292.1081	310.6654	343.0759
98.8497	184.7785	207.6794	229.2178	258.5806
62.7583	129.3102	155.8157	195.4541	195.6949
90.7565	137.5167	148.8166	189.5615	224.7878
92.4098	142.0955	143.3243	175.3780	205.7156
62.2618	110.1159	108.3154	128.3918	151.6979
54.7918	72.9389	69.7607	89.6145	120.0375
24.8789	58.3648	69.5358	76.5516	88.7720
21.2570	63.7878	75.0884	78.2964	84.4967
28.7726	62.7204	61.3969	81.9033	97.0837
22.5679	41.5906	39.5364	57.9563	65.3887
17.1315	47.2470	51.6425	55.5342	58.0956
21.6037	38.2590	25.0555	31.3961	44.7472
14.1799	24.7773	16.9171	24.8576	30.8650
11.8066	24.3758	27.2004	34.3011	37.2119
10.4473	22.2374	24.0919	30.6284	33.3376
1.4530	2.6520	2.5396	3.3148	3.5946
0	0	0	0	0
4.6010	8.3982	8.0421	10.4969	11.3829
5.3275	9.7242	9.3120	12.1543	13.1802

Columns 16 through 20

-3.2361	4.3248	2.1384	10.5493	8.9645
-6.9488	-13.5855	-21.1065	-22.4825	-16.0879
22.8474	43.0276	34.8144	23.9720	24.0379
22.3021	33.4650	66.3516	57.9017	38.0843
-12.2069	-16.3095	-3.5064	-15.3408	-30.7967
-4.8942	-1.0892	-24.9237	-38.1016	-5.0967
99.2346	60.8098	79.9307	109.2277	135.9970
189.9019	166.7508	164.1513	114.7231	118.6813
218.6513	189.2589	146.9302	83.1204	85.7572
236.9630	198.2829	127.0660	80.0098	59.7518
323.8476	275.7831	288.9016	240.1899	217.5697
500.0919	455.5760	458.9728	352.8844	318.9440
542.0432	601.0411	534.4162	413.2673	381.0693
473.6297	549.5753	589.4463	444.4121	419.4603
643.0014	509.4047	589.2027	470.7377	464.4739
877.1757	709.5834	555.5426	452.4654	486.5420
709.5834	847.5107	654.9739	462.9174	483.5505
555.5426	654.9739	804.1633	610.6110	521.7131
452.4654	462.9174	610.6110	674.2720	562.3899
486.5420	483.5505	521.7131	562.3899	655.6409
442.3289	419.0420	438.9075	357.5037	453.7579
412.7575	381.4652	456.1317	348.7722	336.0497
377.0742	373.1288	457.5810	404.0725	399.3041
339.8982	344.7781	458.4838	389.9205	409.6483
261.4702	263.0523	333.4498	261.1606	256.1874

209.9210	191.8047	237.3786	201.0089	208.1954
232.8633	205.0825	255.2733	192.3827	227.7113
203.6064	181.7432	240.4694	170.0120	194.7016
147.2429	131.8821	186.0627	147.6192	144.1428
107.7155	94.8464	143.4343	124.6745	119.5355
77.9069	90.0811	119.7436	83.0718	84.7463
72.5697	67.9349	102.3944	70.0015	69.1416
75.2487	62.9398	104.4180	81.1264	91.3768
72.4871	56.0888	69.3041	45.1433	68.2692
54.6087	46.9640	69.6015	54.2938	61.2277
24.1860	31.3924	56.3284	51.5466	48.1349
9.6454	18.0197	38.2657	36.1876	37.4741
14.2985	13.6247	38.1358	33.8670	38.4566
8.9849	13.2852	36.7553	31.9948	35.5049
-0.4328	1.4180	4.6856	4.3572	4.3176
0	0	0	0	0
-1.3705	4.4905	14.8377	13.7979	13.6725
-1.5869	5.1995	17.1805	15.9765	15.8313

Columns 21 through 25

-0.0005	5.3103	12.8935	15.4054	7.5838
-6.8996	-6.5524	-19.3562	-19.3834	-9.4126
16.6554	8.5245	9.9961	13.6788	21.3475
11.0499	15.5023	32.7887	33.7886	24.0167
-7.4633	4.0216	-23.7605	-31.0623	-18.4964
6.0708	21.2864	-9.7468	-21.2544	-9.6298
87.1615	75.3542	83.1912	79.3613	49.1986
69.6400	54.1597	92.9578	76.6057	55.0861
77.4081	79.5750	75.6352	69.4344	62.6169
103.4084	71.0167	-2.0931	18.1795	25.0805
201.5884	149.6577	140.3750	147.6868	98.8497
273.0499	233.2335	233.4603	250.0523	184.7785
325.6485	290.1750	288.6354	292.1081	207.6794
356.0747	344.5104	335.0294	310.6654	229.2178
434.4923	408.0014	354.3731	343.0759	258.5806
442.3289	412.7575	377.0742	339.8982	261.4702
419.0420	381.4652	373.1288	344.7781	263.0523
438.9075	456.1317	457.5810	458.4838	333.4498
357.5037	348.7722	404.0725	389.9205	261.1606
453.7579	336.0497	399.3041	409.6483	256.1874
539.4084	416.6600	321.4739	341.8563	274.2207
416.6600	493.0858	385.4767	340.0067	283.2016
321.4739	385.4767	470.5939	404.4766	242.1136
341.8563	340.0067	404.4766	463.3960	286.9540
274.2207	283.2016	242.1136	286.9540	316.0608
192.6200	225.8952	214.0845	191.5848	220.4950
234.0815	256.6510	260.7909	249.0131	190.3658
215.3759	246.3480	242.8685	252.9248	206.8864
128.1450	179.2371	180.2628	197.7424	190.8320
105.7375	144.0214	139.4653	151.9677	156.0390

82.0889	101.7668	121.3108	116.9230	102.2037
81.4297	103.0951	105.9099	112.6194	98.7534
76.9762	108.3014	98.5686	115.8171	109.9703
67.0770	70.7615	60.1473	64.6221	68.5314
58.3767	66.1661	62.1164	70.2829	65.0079
34.7236	51.4972	47.7696	56.0336	59.1975
21.4357	33.8315	33.3372	39.3588	41.5656
31.4501	44.4210	35.3119	45.9192	39.0773
26.9838	39.9221	34.9172	43.4169	36.8531
2.5509	4.3890	4.6611	5.3738	4.9541
0	0	0	0	0
8.0777	13.8986	14.7602	17.0171	15.6881
9.3531	16.0931	17.0908	19.7040	18.1652

Columns 26 through 30

8.5833	7.1837	5.3908	8.4062	7.8771
-1.4920	-4.4457	-7.4433	-4.0759	-3.9423
13.3261	-0.3408	-1.5588	9.8031	11.4778
16.4782	17.2504	24.5528	20.0336	22.8013
5.2872	18.3933	-0.1194	-17.5862	-16.6961
17.4044	8.0075	-8.3791	-19.7636	-22.5145
62.3083	54.3983	39.6433	17.0731	1.7740
53.7841	44.1296	21.2919	17.3366	10.9588
53.9184	44.6238	41.2353	40.0373	36.9263
2.1200	3.2766	-8.9991	-7.3550	10.8777
62.7583	90.7565	92.4098	62.2618	54.7918
129.3102	137.5167	142.0955	110.1159	72.9389
155.8157	148.8166	143.3243	108.3154	69.7607
195.4541	189.5615	175.3780	128.3918	89.6145
195.6949	224.7878	205.7156	151.6979	120.0375
209.9210	232.8633	203.6064	147.2429	107.7155
191.8047	205.0825	181.7432	131.8821	94.8464
237.3786	255.2733	240.4694	186.0627	143.4343
201.0089	192.3827	170.0120	147.6192	124.6745
208.1954	227.7113	194.7016	144.1428	119.5355
192.6200	234.0815	215.3759	128.1450	105.7375
225.8952	256.6510	246.3480	179.2371	144.0214
214.0845	260.7909	242.8685	180.2628	139.4653
191.5848	249.0131	252.9248	197.7424	151.9677
220.4950	190.3658	206.8864	190.8320	156.0390
268.7766	196.2217	154.9879	143.3459	117.6393
196.2217	261.3301	229.0355	153.4635	122.5051
154.9879	229.0355	260.7898	191.3366	139.9461
143.3459	153.4635	191.3366	212.4859	154.8370
117.6393	122.5051	139.9461	154.8370	162.1510
90.3108	101.9775	110.4707	93.8870	88.9774
86.7624	87.7252	108.6790	105.6202	68.9104
91.4160	90.0288	114.1518	127.4079	95.2517
75.5901	91.5207	93.9166	79.3888	59.0102
71.6285	80.0767	85.6459	79.0800	58.2294

42.6101	41.3452	58.4897	65.5444	65.0910
29.6720	24.2586	36.6517	43.2603	45.0434
35.3598	32.5013	43.4250	51.1872	43.9452
30.1326	26.1777	38.0238	45.9582	41.1097
3.0050	1.8863	3.8966	5.1124	5.5278
0	0	0	0	0
9.5158	5.9733	12.3393	16.1892	17.5048
11.0183	6.9165	14.2876	18.7454	20.2687

Columns 31 through 35

3.9343	3.7509	3.3790	-0.5265	0.9532
-4.9660	-0.0355	1.6065	0.6061	-0.0136
3.6899	3.6759	6.1042	2.7007	2.9181
16.9354	9.7803	11.9047	6.5348	6.4012
-23.3589	-19.2150	-19.8599	12.7902	-6.1093
-26.2057	-11.7772	1.2004	6.8398	-7.4897
8.5157	15.9529	15.6782	6.2430	9.3096
42.3737	33.3927	16.1710	1.2111	12.9154
25.6168	35.4108	39.2343	17.8190	19.3600
-9.6032	-2.5032	-1.3378	2.4459	-0.1386
24.8789	21.2570	28.7726	22.5679	17.1315
58.3648	63.7878	62.7204	41.5906	47.2470
69.5358	75.0884	61.3969	39.5364	51.6425
76.5516	78.2964	81.9033	57.9563	55.5342
88.7720	84.4967	97.0837	65.3887	58.0956
77.9069	72.5697	75.2487	72.4871	54.6087
90.0811	67.9349	62.9398	56.0888	46.9640
119.7436	102.3944	104.4180	69.3041	69.6015
83.0718	70.0015	81.1264	45.1433	54.2938
84.7463	69.1416	91.3768	68.2692	61.2277
82.0889	81.4297	76.9762	67.0770	58.3767
101.7668	103.0951	108.3014	70.7615	66.1661
121.3108	105.9099	98.5686	60.1473	62.1164
116.9230	112.6194	115.8171	64.6221	70.2829
102.2037	98.7534	109.9703	68.5314	65.0079
90.3108	86.7624	91.4160	75.5901	71.6285
101.9775	87.7252	90.0288	91.5207	80.0767
110.4707	108.6790	114.1518	93.9166	85.6459
93.8870	105.6202	127.4079	79.3888	79.0800
88.9774	68.9104	95.2517	59.0102	58.2294
109.3636	75.7025	57.8057	35.6902	55.8817
75.7025	97.3117	81.6025	39.5992	61.3405
57.8057	81.6025	106.6457	55.4551	59.6714
35.6902	39.5992	55.4551	76.5511	59.8013
55.8817	61.3405	59.6714	59.8013	68.2420
42.5391	39.7998	55.5136	18.0578	26.7182
30.4051	27.2741	42.4542	7.9684	15.3304
35.3974	46.5921	54.6006	24.3254	38.4268
36.1565	43.2969	50.4192	16.2275	31.4177
5.2384	5.1014	6.0116	-0.0961	2.3946

0	0	0	0	0
16.5882	16.1544	19.0366	-0.3042	7.5830
19.2074	18.7051	22.0424	-0.3523	8.7803

Columns 36 through 40

2.8085	1.8205	1.3997	1.4450	0.2599
-0.5472	-0.2063	2.1793	1.5689	0.0240
4.4883	3.7727	3.2492	3.2428	0.5359
9.7290	7.2482	4.9656	5.4837	0.9615
-23.4096	-23.7771	-24.6213	-27.7290	-4.9812
-11.7161	-6.4693	-4.0817	-5.9073	-1.4574
5.6316	5.5379	5.6356	4.5999	0.3948
5.3888	6.9462	8.0203	10.5737	1.9839
15.8903	10.6240	14.0934	13.4731	1.6212
-1.2855	-5.5430	3.9196	0.7765	-0.8882
21.6037	14.1799	11.8066	10.4473	1.4530
38.2590	24.7773	24.3758	22.2374	2.6520
25.0555	16.9171	27.2004	24.0919	2.5396
31.3961	24.8576	34.3011	30.6284	3.3148
44.7472	30.8650	37.2119	33.3376	3.5946
24.1860	9.6454	14.2985	8.9849	-0.4328
31.3924	18.0197	13.6247	13.2852	1.4180
56.3284	38.2657	38.1358	36.7553	4.6856
51.5466	36.1876	33.8670	31.9948	4.3572
48.1349	37.4741	38.4566	35.5049	4.3176
34.7236	21.4357	31.4501	26.9838	2.5509
51.4972	33.8315	44.4210	39.9221	4.3890
47.7696	33.3372	35.3119	34.9172	4.6611
56.0336	39.3588	45.9192	43.4169	5.3738
59.1975	41.5656	39.0773	36.8531	4.9541
42.6101	29.6720	35.3598	30.1326	3.0050
41.3452	24.2586	32.5013	26.1777	1.8863
58.4897	36.6517	43.4250	38.0238	3.8966
65.5444	43.2603	51.1872	45.9582	5.1124
65.0910	45.0434	43.9452	41.1097	5.5278
42.5391	30.4051	35.3974	36.1565	5.2384
39.7998	27.2741	46.5921	43.2969	5.1014
55.5136	42.4542	54.6006	50.4192	6.0116
18.0578	7.9684	24.3254	16.2275	-0.0961
26.7182	15.3304	38.4268	31.4177	2.3946
47.4917	36.7739	31.2100	32.2029	4.9834
36.7739	32.0464	26.1299	28.1515	4.7125
31.2100	26.1299	42.2060	39.2964	4.8504
32.2029	28.1515	39.2964	38.2481	5.2211
4.9834	4.7125	4.8504	5.2211	0.8727
0	0	0	0	0
15.7807	14.9229	15.3596	16.5336	2.7636
18.2724	17.2792	17.7848	19.1442	3.1999

Columns 41 through 43

0	0.8229	0.9529
0	0.0761	0.0881
0	1.6970	1.9649
0	3.0446	3.5253
0	-15.7737	-18.2643
0	-4.6151	-5.3438
0	1.2501	1.4475
0	6.2825	7.2745
0	5.1338	5.9444
0	-2.8125	-3.2566
0	4.6010	5.3275
0	8.3982	9.7242
0	8.0421	9.3120
0	10.4969	12.1543
0	11.3829	13.1802
0	-1.3705	-1.5869
0	4.4905	5.1995
0	14.8377	17.1805
0	13.7979	15.9765
0	13.6725	15.8313
0	8.0777	9.3531
0	13.8986	16.0931
0	14.7602	17.0908
0	17.0171	19.7040
0	15.6881	18.1652
0	9.5158	11.0183
0	5.9733	6.9165
0	12.3393	14.2876
0	16.1892	18.7454
0	17.5048	20.2687
0	16.5882	19.2074
0	16.1544	18.7051
0	19.0366	22.0424
0	-0.3042	-0.3523
0	7.5830	8.7803
0	15.7807	18.2724
0	14.9229	17.2792
0	15.3596	17.7848
0	16.5336	19.1442
0	2.7636	3.1999
0	0	0
0	8.7514	10.1331
0	10.1331	11.7331

```
>> names
```

```
names =
```

```
329×43 char array
```

'Abilene, TX	'
'Akron, OH	'
'Albany, GA	'
'Albany-Troy, NY	'
'Albuquerque, NM	'
'Alexandria, LA	'
'Allentown,Bethlehem, PA-NJ	'
'Alton, Granite City, IL	'
'Altoona, PA	'
'Amarillo, TX	'
'Anaheim-Santa Ana, CA	'
'Anchorage, AK	'
'Anderson, IN	'
'Anderson, SC	'
'Ann Arbor, MI	'
'Anniston, AL	'
'Appleton-Oshkosh-Neenah, WI	'
'Asheville, NC	'
'Athens, GA	'
'Atlanta, GA	'
'Atlantic City, NJ	'
'Augusta, GA-SC	'
'Aurora-Elgin, IL	'
'Austin, TX	'
'Bakersfield, CA	'
'Baltimore, MD	'
'Bangor, ME	'
'Baton Rouge, LA	'
'Battle Creek, MI	'
'Beaumont-Port Arthur, TX	'
'Beaver County, PA	'
'Bellingham, WA	'
'Benton Harbor, MI	'
'Bergen-Passaic, NJ	'
'Billings, MT	'
'Biloxi-Gulfport, MS	'
'Binghampton, NY	'
'Birmingham, AL	'
'Bismarck, ND	'
'Bloomington, IN	'
'Bloomington-Normal, IL	'
'Boise City, ID	'
'Boston, MA	'
'Boulder-Longmont, CO	'
'Bradenton, FL	'
'Brazoria, TX	'
'Bremerton, WA	'
'Bridgeport-Milford, CT	'
'Bristol, CT	'
'Brockton, MA	'
'Brownsville-Harlington, TX	'

```
'Bryan-College Station, TX      '  
'Buffalo, NY                   '  
'Burlington, NC                '  
'Burlington, VT                '  
'Canton, OH                    '  
'Casper, WY                     '  
'Cedar Rapids, IA               '  
'Champaign-Urbana-Rantoul, IL  '  
'Charleston, SC                 '  
'Charleston, WV                 '  
'Charlotte-Gastonia-Rock Hill, NC-SC '  
'Charlottesville, VA           '  
'Chattanooga, TN-GA            '  
'Chicago, IL                    '  
'Chico, CA                      '  
'Cincinnati, OH-KY-IN          '  
'Clarksville-Hopkinsville, TN-KY '  
'Cleveland, OH                  '  
'Colorado Springs, CO           '  
'Columbia, MO                   '  
'Columbia, SC                   '  
'Columbus, GA-AL                '  
'Columbus, OH                   '  
'Corpus Christi, TX             '  
'Cumberland, MD-WV              '  
'Dallas, TX                     '  
'Danbury, CT                    '  
'Danville, VA                   '  
'Davenport-Rock Island-Moline, IA-IL '  
'Dayton-Springfield, OH         '  
'Daytona Beach, FL              '  
'Decatur, IL                    '  
'Denver, CO                     '  
'Des Moines, IA                 '  
'Detroit, MI                    '  
'Dothgan, AL                    '  
'Dubuque, IA                    '  
'Duluth, MN-WI                  '  
'East St. Louis-Belleville, IL  '  
'Eau Claire, WI                 '  
'El Paso, TX                    '  
'Elkhart-Goshen, IN             '  
'Elmira, NY                     '  
'Enid, OK                       '  
'Erie, PA                       '  
'Eugene-Springfield, OR         '  
'Evansville, IN-KY              '  
'Fall River, MA-RI              '  
'Fargo-Moorhead, ND-MN          '  
'Fayetteville, NC               '  
'Fayetteville, AR               '
```

```
'Fitchburg-Leominster, MA'
'Flint, MI'
'Florence, AL'
'Florence, SC'
'Fort Collins-Loveland, CO'
'Fort Lauderdale-Hollywood-Pompano Beach, FL'
'Fort Myers, FL'
'Fort Pierce, FL'
'Fort Smith, AR-OK'
'Fort Walton Beach, FL'
'Fort Wayne, IN'
'Fort Worth-Arlington, TX'
'Fresno, CA'
'Gadsden, AL'
'Gainesville, FL'
'Galveston-Texas City, TX'
'Gary-Hammond, IN'
'Glens Falls, NY'
'Grand Forks, ND'
'Grand Rapids, MI'
'Great Falls, MT'
'Greeley, CO'
'Green Bay, WI'
'Greensboro-Winston-Salem-High Point, NC'
'Greenville-Spartanburg, SC'
'Hagerstown, MD'
'Hamilton-Middletown, OH'
'Harrisburg-Lebanon-Carlisle, PA'
'Hartford, CT'
'Hickory, NC'
'Honolulu, HI'
'Houma-Thibodaux, LA'
'Houston, TX'
'Huntington-Ashland, WV-KY-OH'
'Huntsville, AL'
'Indianapolis, IN'
'Iowa City, IA'
'Jackson, MI'
'Jackson, MS'
'Jacksonville, FL'
'Jacksonville, NC'
'Janesville-Beloit, WI'
'Jersey City, NJ'
'Johnson City-Kingsport-Bristol, TN-VA'
'Johnstown, PA'
'Joliet, IL'
'Joplin, MO'
'Kalamazoo, MI'
'Kankakee, IL'
'Kansas City, KS'
'Kansas City, MO'
```

```
'Kenosha, WI'
'Kileen-Temple, TX'
'Knoxville, TN'
'Kokomo, IN'
'La Crosse, WI'
'Lafayette, IN'
'Lafayette, LA'
'Lake Charles, LA'
'Lake County, IL'
'Lakeland-Winter Haven, FL'
'Lancaster, PA'
'Lansing-East Lansing, MI'
'Laredo, TX'
'Las Cruces, NM'
'Las Vegas, NV'
'Lawrence, KS'
'Lawrence-Haverhill, MA-NH'
'Lawton, OK'
'Lewiston-Auburn, ME'
'Lexington-Fayette, KY'
'Lima, OH'
'Lincoln, NE'
'Little Rock, North Little Rock, AR'
'Longview-Marshall, TX'
'Lorain-Elyria, OH'
'Los Angeles, Long Beach, CA'
'Louisville, KY-IN'
'Lowell, MA-NH'
'Lubbock, TX'
'Lynchburg, VA'
'Macon, Warner Robbins, GA'
'Madison, WI'
'Manchester, NH'
'Mansfield, OH'
'McAllen-Edinburg-Mission, TX'
'Medford, OR'
'Melbourne-Titusville-Palm Bay, FL'
'Memphis, TN-AR-MS'
'Miami-Hialeah, FL'
'Middlesex-Somerset, Hunterdon, NJ'
'Middletown, CT'
'Midland, TX'
'Milwaukee, WI'
'Minneapolis-St. Paul, MN-WI'
'Mobile, AL'
'Modesto, CA'
'Monmouth-Ocean, NJ'
'Monroe, LA'
'Montgomery, AL'
'Muncie, IN'
'Muskegon, MI'
```

```
'Nashua, NH'
'Nashville, TN'
'Nassua-Suffolk, NY'
'New Bedford, MA'
'New Britain, CT'
'New Haven-Meriden, CT'
'New London-Norwich, CT-RI'
'New Orleans, LA'
'New York, NY'
'Newark, NJ'
'Niagara Falls, NY'
'Norfolk-Virginia Beach-Newport News, VA'
'Norwalk, CT'
'Oakland, CA'
'Ocala, FL'
'Odessa, TX'
'Oklahoma City, OK'
'Olympia, WA'
'Omaha, NE-IA'
'Orange County, NY'
'Orlando, FL'
'Owensboro, KY'
'Oxnard-Ventura, CA'
'Panama City, FL'
'Parkerburg-Marietta, WV-OH'
'Pascagoula, MS'
'Pawtucket-Woonsocket-Attleboro, RI-MA'
'Pensacola, FL'
'Peoria, IL'
'Philadelphia, PA-NJ'
'Phoenix, AZ'
'Pine Bluff, AR'
'Pittsburgh, PA'
'Pittsfield, MA'
'Portland, ME'
'Portland, OR'
'Portsmouth-Dover-Rochester, NH-ME'
'Poughkeepsie, NY'
'Providence, RI'
'Provo-Orem, UT'
'Pueblo, CO'
'Racine, WI'
'Raleigh-Durham, NC'
'Reading, PA'
'Redding, CA'
'Reno, NV'
'Richland-Kinnewick-Pasco, WA'
'Richmond-Petersburg, VA'
'Riverside-San Bernardino, CA'
'Roanoke, VA'
'Rochester, MN'
```



```
'Rochester, NY'
'Rockford, IL'
'Sacramento, CA'
'Saginaw-Bay City-Midland, MI'
'St. Cloud, MN'
'St. Joseph, MO'
'St. Louis, MO-IL'
'Salem, OR'
'Salem, MA'
'Salinas-Seaside-Monterey, CA'
'Salt Lake City-Ogden, UT'
'San Angelo, TX'
'San Antonio, TX'
'San Diego, CA'
'San Francisco, CA'
'San Jose, CA'
'Santa Barbara-Santa Maria-Lompoc, CA'
'Santa Cruz, CA'
'Santa Rosa-Petaluma, CA'
'Sarasota, FL'
'Savannah, GA'
'Scranton-Wilkes Barre, PA'
'Seattle, WA'
'Sharon, PA'
'Sheboygan, WI'
'Sherman-Denison, TX'
'Shreveport, LA'
'Sioux City, IA-NE'
'Sioux Falls, SD'
'South Bend-Mishawaka, IN'
'Spokane, WA'
'Springfield, IL'
'Springfield, MA'
'Springfield, MO'
'Stamford, CT'
'State College, PA'
'Steubenville-Weirton, OH-WV'
'Stockton, CA'
'Syracuse, NY'
'Tacoma, WA'
'Tallahassee, FL'
'Tampa-St. Petersburg-Clearwater, FL'
'Terre Haute, IN'
'Texarkana, TX-Texarkana, AR'
'Toledo, OH'
'Topeka, KS'
'Trenton, NJ'
'Tuscon, AZ'
'Tulsa, OK'
'Tuscaloosa, AL'
'Tyler, TX'
```

```

'Utica-Rome, NY
'Vallejo-Fairfield-Napa, CA
'Vancouver, WA
'Victoria, TX
'Vineland-Millville-Bridgeton, NJ
'Visalia-Tulare-Porterville, CA
'Waco, TX
'Washington, DC-MD-VA
'Materbury, CT
'Waterloo-Cedar Falls, IA
'Wausau, WI
'West Palm Beach-Boca Raton-Delray Beach, FL'
'Wheeling, WV-OH
'Wichita, KS
'Wichita Falls, TX
'Williamsport, PA
'Wilmington, DE-NJ-MD
'Wilmington, NC
'Worcester, MA
'Yakima, WA
'York, PA
'Youngstown-Warren, OH
'Yuba City, CA

```

```
>> CovRatings = cov(ratings)
```

```
CovRatings =
```

```
1.0e+07 *
```

```
Columns 1 through 5
```

0.0015	0.0111	0.0026	0.0008	0.0014
0.0111	0.5689	0.1084	0.0114	0.0941
0.0026	0.1084	0.1006	0.0109	0.0685
0.0008	0.0114	0.0109	0.0128	0.0149
0.0014	0.0941	0.0685	0.0149	0.2106
0.0003	0.0151	0.0158	0.0009	0.0156
0.0127	0.4967	0.4031	0.0646	0.3131
0.0021	0.0814	0.0264	0.0099	0.0428
-0.0013	0.0697	0.0075	0.0101	0.0093

```
Columns 6 through 9
```

0.0003	0.0127	0.0021	-0.0013
0.0151	0.4967	0.0814	0.0697
0.0158	0.4031	0.0264	0.0075
0.0009	0.0646	0.0099	0.0101
0.0156	0.3131	0.0428	0.0093
0.0103	0.0556	0.0020	0.0042
0.0556	2.1551	0.1420	0.0381

0.0020	0.1420	0.0653	0.0152
0.0042	0.0381	0.0152	0.1176

```
>> EigCategories = eig(categories)
```

```
Error using eig
```

```
Invalid data type. Input matrix must be double or  
single.
```

```
>> EigCategories = eig(CovCategories)
```

```
EigCategories =
```

```
1.0e+03 *
```

```
-0.0000  
-0.0000  
-0.0000  
-0.0000  
0.0000  
0.0000  
0.0127  
0.0181  
0.1485  
0.4477  
0.5110  
1.2028  
2.0951  
6.1752
```

```
>> EigNames = eig(CovNames)
```

```
EigNames =
```

```
1.0e+03 *
```

```
-0.0000  
0.0000  
0.0000  
0.0000  
0.0001  
0.0005  
0.0008  
0.0039  
0.0049  
0.0064  
0.0088  
0.0136  
0.0190  
0.0254  
0.0335  
0.0370
```

```
0.0383
0.0457
0.0505
0.0658
0.0736
0.0872
0.0999
0.1064
0.1214
0.1380
0.1548
0.1880
0.2063
0.2203
0.2612
0.2862
0.3530
0.4011
0.4880
0.5138
0.6934
0.8227
0.9283
1.2821
1.3792
1.8206
7.0513
```

```
>> EigRatings1 = eig(ratings)
Error using eig
Input matrix must be square.
```

```
>> EigRatings = eig(CovRatings)
```

```
EigRatings =
```

```
1.0e+07 *

0.0011
0.0067
0.0093
0.0241
0.0478
0.1076
0.1638
0.4408
2.4414
```

```
>> CoeffCategories = pca(categories)
Error using pca (line 168)
The value of X must be a numeric array.
```

```
>> Coeff = pca(names)
```

```
Error using pca (line 168)
```

```
The value of X must be a numeric array.
```

```
>> Coeff = pca(ratings)
```

```
Coeff =
```

```
Columns 1 through 5
```

0.0064	0.0155	-0.0067	-0.0263	0.0163
0.2691	0.9372	-0.0826	-0.1778	-0.0838
0.1783	-0.0205	0.0278	-0.0266	-0.1591
0.0281	-0.0109	0.0376	0.0990	0.1160
0.1493	0.0188	0.9715	-0.0384	-0.1466
0.0252	-0.0014	0.0415	0.0216	-0.1063
0.9309	-0.2823	-0.1510	0.0278	0.0087
0.0698	0.1038	0.1496	0.0690	0.9543
0.0251	0.1734	0.0127	0.9745	-0.1022

```
Columns 6 through 9
```

-0.0012	0.0814	0.0421	0.9951
-0.0486	0.0267	0.0121	-0.0229
0.9295	0.1371	-0.2414	0.0014
-0.0540	0.9448	0.2668	-0.0877
-0.0922	-0.0135	-0.0415	0.0094
0.2532	-0.2412	0.9292	-0.0169
-0.1676	-0.0430	0.0159	0.0006
0.1733	-0.1271	0.0188	-0.0050
0.0052	-0.0702	-0.0544	0.0327

```
>> Coeff([1:2],:)
```

```
ans =
```

```
Columns 1 through 5
```

0.0064	0.0155	-0.0067	-0.0263	0.0163
0.2691	0.9372	-0.0826	-0.1778	-0.0838

```
Columns 6 through 9
```

-0.0012	0.0814	0.0421	0.9951
-0.0486	0.0267	0.0121	-0.0229

```
>> Coeff(:,[1:2])
```

```
ans =
```

0.0064	0.0155
0.2691	0.9372
0.1783	-0.0205
0.0281	-0.0109
0.1493	0.0188
0.0252	-0.0014
0.9309	-0.2823
0.0698	0.1038
0.0251	0.1734

```
>> PComp = Coeff(:,[1:2])
```

```
PComp =
```

0.0064	0.0155
0.2691	0.9372
0.1783	-0.0205
0.0281	-0.0109
0.1493	0.0188
0.0252	-0.0014
0.9309	-0.2823
0.0698	0.1038
0.0251	0.1734

```
>> mean names
```

```
ans =
```

```
106.4000
```

```
>> mean ratings
```

```
ans =
```

```
108.5714
```

```
>> mean categories
```

```
ans =
```

```
106.2000
```

```
>> mean(ratings)
```

```
ans =
```

```
1.0e+03 *
```

```
Columns 1 through 5
```

0.5387	8.3466	1.1857	0.9611	4.2101
--------	--------	--------	--------	--------

Columns 6 through 9

2.8149	3.1509	1.8460	5.5254
--------	--------	--------	--------

```
>> mean(names)
```

ans =

Columns 1 through 5

74.9483	105.6717	107.1763	102.8602	100.0760
---------	----------	----------	----------	----------

Columns 6 through 10

97.9179	91.6049	85.3860	82.3435	78.6383
---------	---------	---------	---------	---------

Columns 11 through 15

70.1429	64.2553	62.2918	58.1976	53.3739
---------	---------	---------	---------	---------

Columns 16 through 20

50.9149	49.6201	46.9635	43.9514	43.4924
---------	---------	---------	---------	---------

Columns 21 through 25

41.6383	40.5167	40.2979	40.0578	37.7933
---------	---------	---------	---------	---------

Columns 26 through 30

36.4316	36.7204	36.2462	35.1307	34.4529
---------	---------	---------	---------	---------

Columns 31 through 35

33.9088	33.7812	33.8419	33.3131	33.2736
---------	---------	---------	---------	---------

Columns 36 through 40

32.8936	32.5957	32.7112	32.6444	32.0729
---------	---------	---------	---------	---------

Columns 41 through 43

32.0000	32.2310	32.2675
---------	---------	---------

```
>> mean(categories)
```

ans =

Columns 1 through 5

103.8889 106.8889 107.1111 109.8889 97.2222

Columns 6 through 10

92.0000 81.8889 65.4444 68.0000 47.8889

Columns 11 through 14

41.3333 40.1111 40.7778 40.6667

>> mean(ratings)

ans =

1.0e+03 *

0.5387 8.3466 1.1857 0.9611 4.2101 2.8149 3.1509 1.8460 ✓
5.5254

>> mean(names)

ans =

Columns 1 through 9

74.9483 105.6717 107.1763 102.8602 100.0760 97.9179 91.6049 85.3860 ✓
82.3435

Columns 10 through 18

78.6383 70.1429 64.2553 62.2918 58.1976 53.3739 50.9149 49.6201 ✓
46.9635

Columns 19 through 27

43.9514 43.4924 41.6383 40.5167 40.2979 40.0578 37.7933 36.4316 ✓
36.7204

Columns 28 through 36

36.2462 35.1307 34.4529 33.9088 33.7812 33.8419 33.3131 33.2736 ✓
32.8936

Columns 37 through 43

32.5957 32.7112 32.6444 32.0729 32.0000 32.2310 32.2675

>> mean(categories)

ans =

Columns 1 through 9

103.8889	106.8889	107.1111	109.8889	97.2222	92.0000	81.8889	65.4444	✓
68.0000								

Columns 10 through 14

47.8889	41.3333	40.1111	40.7778	40.6667
---------	---------	---------	---------	---------

```
>> biplot(Coeff)
```

```
Error using biplot (line 92)
```

```
COEFS must have 2 or 3 columns.
```

```
>>
```